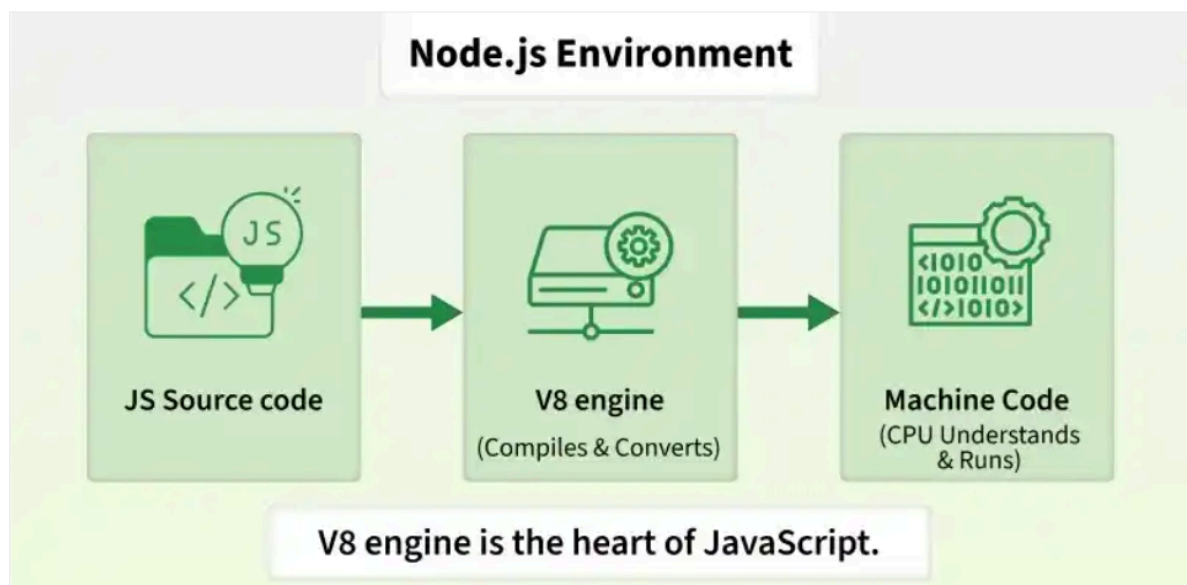


Introduction To Node JS -

Node.js is an open-source, cross-platform JavaScript runtime built on Chrome's V8 engine. It enables developers to run JavaScript outside the browser to build fast, scalable server-side applications.

- JavaScript was initially a frontend-only language, and Node.js (2009) enabled backend development as well.
- Non-blocking, event-driven architecture for high performance.
- Supports the creation of REST APIs, real-time applications, and microservices.
- Comes with a rich library of modules through npm (Node Package Manager).

*NOTE - Node.js is **not a programming language** and it is **not a framework**. It is a runtime environment for executing JavaScript code without a browser.*



What is the V8 Engine?

The V8 engine is Google's open-source JavaScript engine, used by Chrome and Node.js. Without V8 engine, Node.js would not be able to directly execute JavaScript.

It compiles JavaScript to native machine code for fast execution.

- **Origin:** Developed by Google for Chrome in 2008
- **Integration:** Node.js uses V8 to provide a JavaScript runtime on the server
- **Features:** Just-In-Time compilation, ES6+ support

Creation of Node.js -

In **2009**, **Ryan Dahl** introduced Node.js.

His goal was to create a way to use JavaScript for **server-side programming** with an efficient, event-driven architecture.

Node.js used Google's **V8 JavaScript engine** as its JavaScript execution engine.

Installation of Node.js -

Go to official website - <https://nodejs.org/en/download/current>

Select the latest version and download the installer according to your operating system. Run the installer and keep the default options.

Verify the installation -

After installation, open:

Windows - Command Prompt / PowerShell on VS code
macOS/Linux - Terminal

Run: node -v

You should get something like:
v26.8.1

The exact version will depend on the current LTS release.

First Node.js program -

Create a folder and open it in VS Code.

Create app.js file.

Write: `console.log("Hello from Node.js!");`

Now open the terminal and run:

`node app.js`

Output:

Hello from Node.js!

That's your first Node program. Here, you will notice that javascript code ran without any browser because it ran with the help of Node.js. It provided the environment(V8 engine) to run the code.

Node provides its own environment and APIs.

For example, Node can interact with the filesystem:

```
const fs = require("fs");  
fs.writeFileSync("hello.txt", "Hello Node!");
```

Run: node app.js

A file called: **hello.txt** will be created.

NOTE - JavaScript is now interacting with the computer rather than just the webpage.

Node.js is widely used for:

- Web servers
- REST APIs
- Real-time applications
- Microservices
- Command-line applications
- Backend applications
- Streaming applications

Features of Node.js -

1. JavaScript Runtime -

Node.js allows us to execute JavaScript outside the browser.

```
const name = "Ekta";  
console.log(`Hello ${name}`);
```

Run: **node app.js**

2. Built on V8 Engine -

Node.js uses Google's **V8 JavaScript engine** to execute JavaScript.

V8 converts JavaScript into machine-level instructions efficiently.

3. Asynchronous -

Node.js supports **asynchronous operations**.

For example, when Node.js needs to read a file, it doesn't necessarily have to stop the entire application while waiting for the file operation to finish. The program can continue doing other work while the file operation is being completed.

4. Non-Blocking I/O -

I/O = Input/Output

Examples:

- Reading files
- Writing files
- Database operations
- Network requests

Node.js uses a non-blocking approach for many I/O operations.

Blocking approach

Task A



Wait



Task B

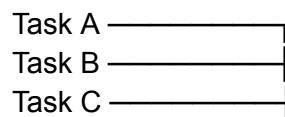


Wait



Task C

Non-blocking approach



Results handled
when available

This makes Node.js particularly useful for applications that handle many I/O operations.

5. Event-Driven -

Node.js follows an **event-driven architecture**.

An event can be something like:

- Request received
- File reading completed
- User connected

- Database operation completed

6. Single-Threaded JavaScript Execution -

Node.js executes JavaScript code using a **single main JavaScript thread**. However, this does **not** mean Node.js can only perform one operation at a time. Node.js can handle asynchronous I/O using its event loop and underlying system mechanisms.

7. Fast and Scalable -

Because of its event-driven and non-blocking architecture, Node.js can efficiently handle applications with many concurrent I/O operations.

It is commonly used for:

- Chat applications
- APIs
- Streaming applications
- Real-time applications

8. npm Ecosystem -

npm = Node Package Manager

Node.js has access to a huge ecosystem of reusable packages through **npm (Node Package Manager)**.

It is used to:

- Install packages
- Manage dependencies
- Run scripts
- Manage project configuration

Run on terminal: `npm -v`

You should also get a version number.

REPL in Node.js -

REPL = Read-Eval-Print Loop

Node.js provides a built-in interactive JavaScript environment called the **Node.js REPL**. It allows us to execute JavaScript code directly from the terminal without creating a `.js` file.

Starting REPL -

Open the terminal and type:

```
node
```

You will see something similar to:

```
>
```

Now you can write JavaScript:

```
> 10 + 20  
30
```

```
> "Hello".toUpperCase()  
'HELLO'
```

```
> let x = 100  
undefined
```

```
> x + 50  
150
```

R — Read

Reads the JavaScript expression entered by the user.

E — Eval

Evaluates/executes the expression.

P — Print

Prints the result.

L — Loop

Waiting for the next command.

REPL is useful for:

- Learning JavaScript
- Testing small pieces of code
- Debugging
- Experimenting with Node.js APIs
- Quickly checking syntax or expressions

Example -

```
> const name = "node"
undefined
> name
'node'
> name.length
4
```

Exit REPL -

You can exit using: `.exit` or press: **Ctrl + C** twice.

Global Objects -

Global objects are objects/functions/values that are available throughout a Node.js application without explicitly importing them.

Example 1:

```
console.log("Hello");
```

We don't need to import the `console` first.

Example 2:

```
setTimeout(() => {  
  console.log("Hello");  
}, 2000);
```

`setTimeout()` can be used directly.

Example 3:

```
console.log(process.version);
```

The `process` object provides information and control over the current Node.js process. This displays the Node.js version.

Understanding the JavaScript Event Loop

Call Stack • Web APIs • Callback Queue • Microtask Queue • The Event Loop

Before we start

This is one of the most misunderstood topics in JavaScript — and also one of the most rewarding to truly understand. Once the Event Loop “clicks,” async code stops feeling like magic and starts feeling like simple, predictable machinery. Take it one section at a time, run the examples yourself, and you’ll get there. Let’s go!

1. Core Concepts: The Call Stack & the Global Execution Context

1.1 What is the Global Execution Context (GEC)?

The moment a JavaScript file starts running, the JS engine creates a special container called the **Global Execution Context (GEC)**. Think of it as the “main stage” of your program — it is created first, and it is destroyed last. Every other piece of code runs *inside* it, directly or indirectly.

1.2 What is the Call Stack?

The **Call Stack** is a simple data structure that keeps track of *where the program currently is*. It works on a **LIFO** principle — *Last In, First Out*. Whenever a function is called, JavaScript pushes a new frame onto the stack. When that function finishes (returns), its frame is popped off.

Mental Model

Imagine a stack of plates. The GEC is the table the stack sits on. Every function call adds a plate on top; every function that finishes removes the plate from the top. You can only ever look at (execute) the very top plate.

1.3 Step-by-Step Trace: A Simple Synchronous Example

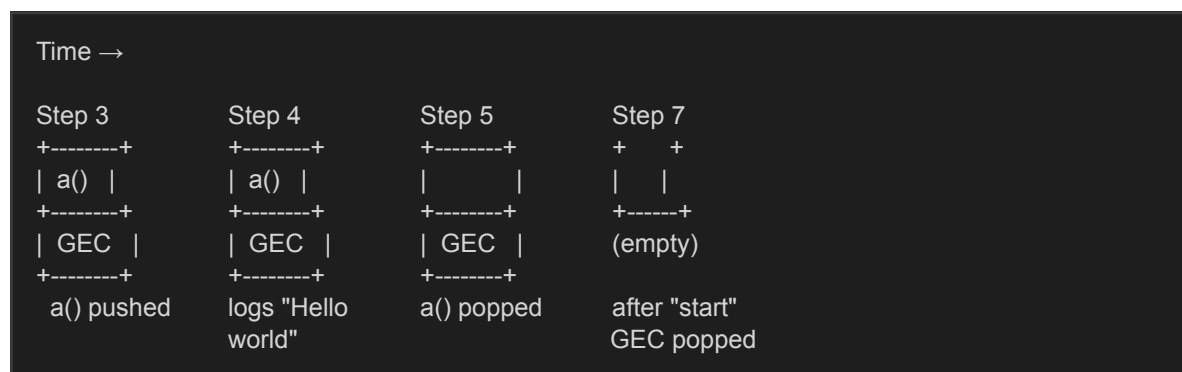
Let’s trace exactly what the JS engine does with this code:

```
function a() {  
  console.log("Hello world");  
}  
  
a();  
console.log("start");
```

Execution trace

1. JS engine starts the file → a Global Execution Context (GEC) is created and pushed onto the Call Stack. Stack: [GEC]
2. Function a() is declared (hoisted) but not run yet — nothing happens on the stack for a declaration.
3. Line a() executes → a new Function Execution Context for a() is created and pushed on top of the GEC. Stack: [GEC, a()]
4. Inside a(), console.log("Hello world") runs → "Hello world" is printed immediately.
5. a() has nothing left to do → its execution context is popped off the stack. Stack: [GEC]
6. Back in the GEC, console.log("start") runs → "start" is printed.
7. The file has no more code → the GEC itself is popped off. Stack: [] (empty).

Visualizing the stack over time



// Output:

Hello world
start

⚠ Common Misconception

"Functions run in the order they are written." Not quite — functions run in the order they are CALLED and INVOKED on the stack, which is why a() (called first) fully finishes before console.log("start") even though both lines look sequential. Order comes from the stack, not just the source order.

2. Web APIs / Browser APIs — Why JavaScript Needs Help

2.1 JavaScript itself is single-threaded

Here's the key fact that makes the Event Loop necessary: **the JavaScript engine (like V8) only has ONE Call Stack and can only do ONE thing at a time**. It has *no built-in ability* to “wait” for something (like a timer or a network request) without freezing the entire program.

2.2 So who handles waiting?

JavaScript engines don't run in a vacuum — they run **inside an environment**, such as a **browser** (Chrome, Firefox) or **Node.js**. These environments provide extra capabilities that plain JS does not have, called **Web APIs** in the browser (Node has its own equivalent APIs, e.g. libuv-backed timers and I/O). These include:

- **setTimeout / setInterval** — timers that count down in the background
- **fetch / XMLHttpRequest** — network requests handled outside the JS thread
- **DOM Events** — click, scroll, keypress listeners
- **Geolocation, localStorage, requestAnimationFrame**, and more

Key Idea

The browser (or Node) essentially says: “Don't worry, JS — I'll handle the waiting for you. You keep running other code, and I'll let you know the moment the timer finishes / the data arrives.” This is what makes JavaScript feel asynchronous even though the engine itself is single-threaded.

2.3 Step-by-Step Trace: Introducing setTimeout

```
console.log("start");

setTimeout(() => {
  console.log("Async Task");
}, 2000);

console.log("end");
```

Execution trace

1. GEC is created and pushed onto the Call Stack.
2. `console.log("start")` runs → "start" printed immediately.
3. `setTimeout(...)` is called. JS does NOT run the timer itself — it hands the callback function and the 2000ms delay off to the Web API environment (the

browser), and immediately continues to the next line. The `setTimeout` call itself is popped off the stack right away.

4. `console.log("end")` runs → "end" printed immediately.
5. The main script finishes → GEC is popped → Call Stack is empty.
6. Meanwhile, the browser's internal timer counts down in the background (this does NOT block the stack).
7. After ~2000ms, the timer completes. The callback `() => console.log("Async Task")` is moved into the Callback Queue (Task Queue), waiting for its turn.
8. The Event Loop (explained in Section 4) checks: "Is the Call Stack empty?"
Yes → it pushes the callback from the queue onto the stack to run it.
9. "Async Task" is finally printed — roughly 2 seconds after the script started, but AFTER "start" and "end".

// Output:

start

end

Async Task (≈ 2 seconds later)

⚠ Common Misconception

"`setTimeout(fn, 2000)` guarantees `fn` runs at exactly 2000ms." Not true — 2000ms is the MINIMUM delay before the callback is eligible to move to the queue. If the Call Stack is busy with other code when the timer finishes, the callback has to wait its turn. The delay is a floor, not a promise.

3. Callback Queue vs. Microtask Queue

Once a Web API finishes its job (timer ends, data arrives, etc.), it doesn't throw its callback directly onto the Call Stack. Instead, it places the callback into a *waiting line* — and JavaScript actually has **two** such waiting lines, with different priority.

3.1 The Callback Queue (a.k.a. Macrotask / Task Queue)

Holds callbacks from:

- `setTimeout` / `setInterval`
- DOM events (click, keydown, etc.)
- `setImmediate` (Node.js)
- I/O callbacks in Node.js

3.2 The Microtask Queue (Priority Queue)

Holds callbacks from:

- **Promise**.then() / .catch() / .finally()
- **async/await** (an awaited call resumes as a microtask)
- queueMicrotask()
- **MutationObserver** callbacks



The Golden Rule

The Microtask Queue always has priority over the Callback (Macrotask) Queue. After every single task finishes running on the Call Stack, the Event Loop must FULLY drain the entire Microtask Queue — including any NEW microtasks added along the way — before it is allowed to pick even one task from the Callback Queue.

3.3 Visual Comparison

Feature	Callback Queue (Macrotask)	Microtask Queue
Examples	setTimeout, setInterval, DOM events	Promise.then/catch/finally, async/await, queueMicrotask
Priority	Lower — runs after microtasks are empty	Higher — always runs first
How many run per Event Loop tick	Only ONE macrotask is taken per loop cycle	ALL microtasks run, until the queue is fully empty
Nickname	“Task Queue”	“Priority Queue”

4. The Event Loop — The Bridge That Connects Everything

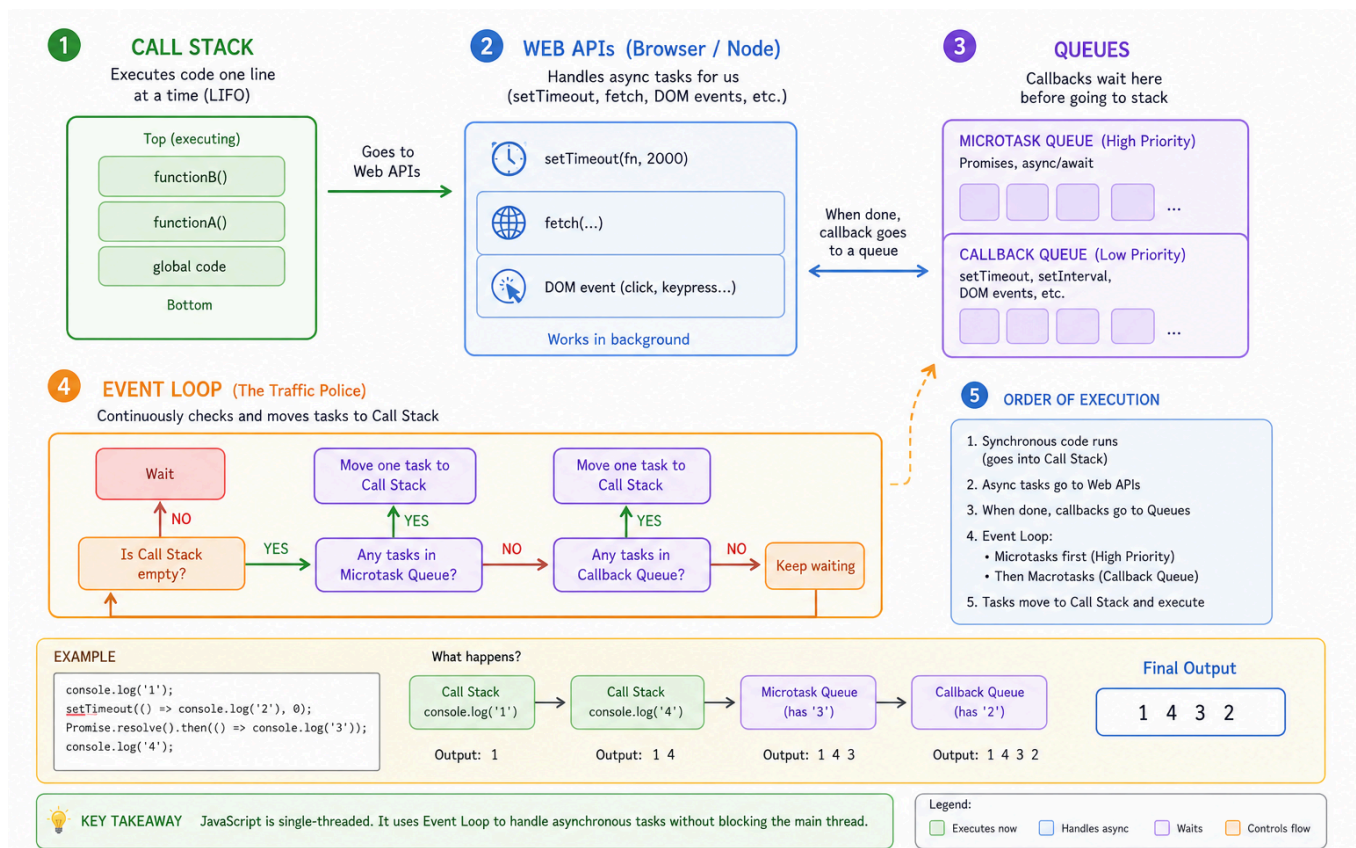
The **Event Loop** is not a queue itself — it's a constantly-running *watchman / traffic controller*. Its ONE job, repeated forever, is beautifully simple:



The Event Loop's Job (in plain English)

“Is the Call Stack empty? If yes — first, run every single microtask waiting in the Microtask Queue (fully draining it). Then, take ONE task from the Callback Queue and push it onto the Call Stack. Repeat forever.”

4.1 The full picture



4.2 Why the Event Loop matters

Without the Event Loop, JavaScript would have no way to know *when* it's safe to run a queued callback. It is the mechanism that lets a single-threaded language handle thousands of timers, network requests, and UI events without ever blocking the page — as long as the Call Stack is respected: **never run a queued callback while the stack is still busy.**

5. Full Code Walkthroughs

5.1 Walkthrough A — Basic Async Task

```
console.log("1: Sync start");

setTimeout(() => {
  console.log("2: Timeout callback");
}, 0);

Promise.resolve().then(() => {
  console.log("3: Promise callback");
});

console.log("4: Sync end");
```

Trace

1. "1: Sync start" logs immediately (synchronous, runs on the Call Stack).
2. `setTimeout` hands its callback to the Web API. Even with a 0ms delay, it still must go through the Callback Queue — it cannot run immediately.
3. `Promise.resolve().then(...)` schedules its callback into the Microtask Queue.
4. "4: Sync end" logs immediately (still synchronous code running).
5. Call Stack is now empty. Event Loop checks the Microtask Queue FIRST → runs it → "3: Promise callback" logs.
6. The Microtask Queue is now empty. Event Loop moves to the Callback Queue → runs the timeout callback → "2: Timeout callback" logs.

// Output: 1: Sync start

// Output: 4: Sync end

// Output: 3: Promise callback

// Output: 2: Timeout callback

⚠ Common Misconception

"`setTimeout(fn, 0)` runs instantly, right after the current line." False — even a 0ms `setTimeout` must go through the Web API and the Callback Queue, so it ALWAYS runs after the synchronous code AND after all microtasks, no matter how short the delay is.

5.2 Walkthrough B — Priority Battle: `setTimeout` vs. `fetch` (Promise)

This example shows exactly why the Microtask Queue is called the "Priority Queue" — even when a Promise resolves later than a timer's Web API work theoretically starts, the microtask still cuts in line ahead of the queued macrotask.

```
console.log("A: script start");

setTimeout(() => {
  console.log("B: setTimeout (macrotask)");
}, 0);

fetch("https://example.com/data")
  .then(() => console.log("C: fetch (microtask)"));

Promise.resolve().then(() => console.log("D: promise (microtask)"));

console.log("E: script end");
```

Trace

1. "A: script start" logs — synchronous.
2. `setTimeout` registers its callback with the Web API and returns immediately (callback destined for the Callback / Macrotask Queue once the timer elapses).
3. `fetch()` kicks off a network request via the Web API (this takes real time) and immediately returns a pending Promise; `.then()` registers a microtask to run once that promise resolves.
4. `Promise.resolve().then(...)` is already resolved, so its callback is queued into the Microtask Queue right away.
5. "E: script end" logs — synchronous code finishes, Call Stack empties.
6. Event Loop drains the Microtask Queue completely before touching the macrotask queue → "D: promise (microtask)" logs first (it was ready immediately).
7. The 0ms `setTimeout` callback is likely ready in the Callback Queue around this time too, but the Event Loop must fully empty the Microtask Queue on every pass before taking even one macrotask → "B: setTimeout (macrotask)" logs next (assuming the network hasn't responded yet).
8. Once the network response for `fetch()` actually arrives (this could be milliseconds or longer, and is NOT guaranteed to beat the timer), its microtask joins the Microtask Queue and is drained before any further macrotasks → "C: fetch (microtask)" logs.

// Output: A: script start

// Output: E: script end

// Output: D: promise (microtask)

// Output: B: setTimeout (macrotask) <-- typical order, network permitting

// **Output:** C: fetch (microtask)

<-- arrives whenever the network responds

Why does D beat B, but C's timing can vary?

D (Promise.resolve()) is already resolved the instant it's created — so its microtask is ready to run on the very next Event Loop check, ahead of ANY macrotask. B (setTimeout) is a macrotask, so it must wait for the Microtask Queue to be fully empty first. C (fetch) depends on an actual network round-trip — its microtask can only be queued once the response arrives, which is often (but not always) slower than a 0ms timer. The rule that never changes: whenever the microtask queue has something in it, it is drained completely before the next macrotask runs.

Quick-Reference Summary

- **Call Stack** = single-threaded, LIFO, runs synchronous code right now.
- **Web APIs** (browser/Node) = handle waiting (timers, network, DOM) OUTSIDE the JS thread.
- **Callback / Macrotask Queue** = holds setTimeout, DOM event, I/O callbacks. One task per Event Loop cycle.
- **Microtask Queue** = holds Promise/async-await callbacks. ALWAYS fully drained before the next macrotask.
- **Event Loop** = the bridge: “Is the stack empty? Drain microtasks. Then take one macrotask.” Repeat forever.

You made it!

If this is your first real pass at the Event Loop, don't worry if it takes a couple of re-reads and a few `console.log` experiments in your own browser console to fully click. Every professional JS developer had this exact same “wait, WHY did that print in that order?!” moment at some point. Trace the examples by hand, predict the output before running the code, and you'll master this faster than you think. Happy coding!

1. Modules in Node.js

A **module** is simply a **separate file containing some code that we can reuse in another file**.

Instead of writing everything in one huge file:

```
app.js
├── user code
├── database code
├── authentication code
├── file handling code
└── utility code
```

we can divide it:

```
project/
├── app.js
├── user.js
├── database.js
└── utils.js
```

Each file can act as a **module**.

Why use modules?

- Code becomes easier to understand
- Code can be reused
- Large applications can be divided into smaller parts
- Easier debugging and maintenance

1.1 CommonJS Modules

CommonJS (CJS) is the traditional module system used by Node.js. It uses:

require() to import modules

module.exports to export them. **Example -**

Create: **math.js**

```
function add(a, b) {  
  return a + b;  
}  
  
function multiply(a, b) {  
  return a * b;  
}  
  
module.exports = {  
  add,  
  multiply  
};
```

Now create:

app.js

```
const m = require("./math");  
  
console.log(m.add(10, 20));  
console.log(m.multiply(5, 4));
```

Output:

30
20

What happened?

math.js
|
| module.exports
↓
app.js
|
| require("./math")
↓
Gets the functions

Exporting a single value -

You can also do:

module.exports = add;

Then:

```
const add = require("./math");
```

```
console.log(add(10, 20));
```

1.2 ES Modules

ES Modules (ESM) is the modern JavaScript module system.

It uses: `export` and `import` instead of `module.exports` and `require()`.

math.js

```
export function add(a, b) {  
  return a + b;  
}  
  
export function multiply(a, b) {  
  return a * b;  
}
```

app.js

```
import { add, multiply } from "./math.js";  
  
console.log(add(10, 20));  
console.log(multiply(5, 4));
```

CommonJS vs ES Modules -

CommonJS	ES Modules
Older/traditional Node.js system	Modern JavaScript standard
<code>require()</code>	<code>import</code>
<code>module.exports</code>	<code>export</code>
<code>const fs = require("fs")</code>	<code>import fs from "fs"</code>
Common in older Node projects	Common in modern projects

1.3 Built-in Modules in Node.js

Node.js comes with many modules already. You don't have to install them using npm.

Some important ones are:

fs → File System
path → File and folder paths
http → Create HTTP servers
os → Operating system information
events → Event handling
url → Work with URLs

1.4 File System Module — fs

The **File System module (fs)** allows Node.js to interact with files and folders.

You can:

- Create files
- Read files
- Write files
- Update files
- Delete files
- Rename files
- Create directories
- Read directories

Import it using CommonJS:

```
const fs = require("fs");
```

1. Reading a File

Suppose we have a file:

student.txt

Hello Students!

Welcome to Node.js.

We can read it using:

```
const fs = require("fs");

fs.readFile("student.txt", "utf8", (err, data) => {

    if (err) {
        console.log(err);
        return;
    }

    console.log(data);
});
```

Output:

Hello Students!

Welcome to Node.js.

Understand the parameters:

Here, `fs` is the file system module and `require()` means: **"Give me this module so I can use it in my program."**

`fs.readFile("student.txt", "utf8", callback);`

`student.txt` → Which file?

`utf8` → How should data be decoded?

`(err, data)` is a callback function which means - Once the file reading is finished, execute this function. Here, `err` stands for **error**. If something goes wrong while reading the file, Node.js puts the error information inside `err`. `data` contains the **contents of the file** if the file was successfully read.

2. Writing to a File -

```
const fs = require("fs");
fs.writeFile(
    "student.txt",
    "Welcome to Node.js!",
    (err) => {
        if (err) {
            console.log(err);
            return;
        }
        console.log("File written successfully");
    })
```

```
}  
);
```

If `student.txt` doesn't exist, Node.js can create it. If it already exists, `writeFile()` **replaces its contents**.

3. Appending to a File -

If you don't want to replace existing content:

```
const fs = require("fs");  
  
fs.appendFile(  
  "student.txt",  
  "\nThis is a new line.",  
  (err) => {  
  
    if (err) {  
      console.log(err);  
      return;  
    }  
    console.log("Data added");  
  }  
);
```

Now the file becomes:
Welcome to Node.js!
This is a new line.

4. Deleting a File -

```
const fs = require("fs");  
  
fs.unlink("student.txt", (err) => {  
  
  if (err) {  
    console.log(err);  
    return;  
  }  
})
```

```
    console.log("File deleted");  
  });
```

`unlink()` is commonly used to delete a file.

5. Creating a Folder -

```
const fs = require("fs");  
fs.mkdir("students", (err) => {  
  if (err) {  
    console.log(err);  
    return;  
  }  
  console.log("Folder created");  
});
```

6. Reading a Folder -

```
const fs = require("fs");  
fs.readdir("students", (err, files) => {  
  
  if (err) {  
    console.log(err);  
    return;  
  }  
  console.log(files);  
});
```

What if you want the files in the current folder?

```
const fs = require("fs");
fs.readdir(".", (err, files) => {

  if (err) {
    console.log(err);
    return;
  }
  console.log(files);
});
```

Here "." means **current folder**.

So you'll get something like:

```
[ 'app.js', 'student.txt', 'students' ]
```

Path Module in Node.js

The **Path module** is a built-in Node.js module that helps us **work with file and folder paths**.

For example:

```
C:\Users\Ekta\NodeProject\files\student.txt
```

A path tells us **where a file or folder is located**.

Node.js provides the **path** module so that we can easily create, analyze, and manipulate these paths.

Import the Path Module

With CommonJS:

```
const path = require("path");
```

Here:

- `require("path")` → loads Node.js's built-in Path module
- `path` → variable through which we access its functions

Now we can use:

```
path.join()
path.basename()
path.dirname()
path.extname()
path.parse()
```

Why do we need the Path Module?

If we want to create this path:

files/student/data.txt

We could write:

```
const filePath = "files/student/data.txt";
```

But this can become problematic when working across different operating systems because path separators can differ.

Instead, we can use:

```
const path = require("path");
```

```
const filePath = path.join("files", "student", "data.txt");
```

```
console.log(filePath);
```

Node.js creates the appropriate path for the operating system.

path.join() -

This is one of the **most important Path methods**.

```
const path = require("path");
```

```
const filePath = path.join("students", "student", "data.txt");
```

```
console.log(filePath);
```

Output:

students\student\data.txt

path.join() automatically handles the correct separator.

So, **path.join()** says:

"Node.js, join these folder/file names and create one proper path for me."

path.basename() -

basename() gives us the **last part of a path**, usually the filename.

```
const path = require("path");  
  
const filePath = "/students/student/data.txt";  
  
console.log(path.basename(filePath));
```

Output:

data.txt

path.dirname() -

dirname() gives us the **directory/folder portion** of a path.

```
const path = require("path");  
  
const filePath = "/students/data/student.txt";  
  
console.log(path.dirname(filePath));
```

Output:

/students/data

path.extname() -

extname() gives us the **file extension**.

```
const path = require("path");  
  
console.log(path.extname("student.txt"));
```

Output:

.txt

Another example:

```
console.log(path.extname("photo.jpg"));
```

Output:

.jpg

path.parse() -

`parse()` breaks a path into different pieces.

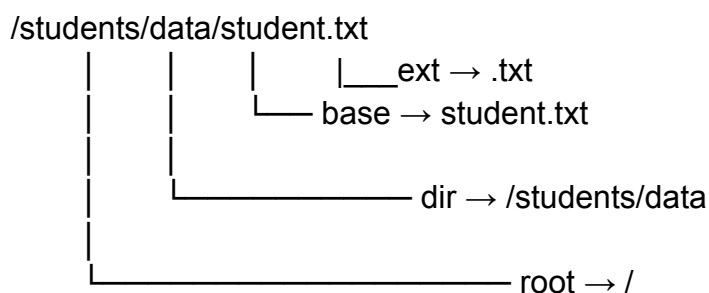
```
const path = require("path");
```

```
const result = path.parse("/students/data/student.txt");
```

```
console.log(result);
```

You'll get an object similar to:

```
{
  root: "/",
  dir: "/students/data",
  base: "student.txt",
  ext: ".txt",
  name: "student"
}
```



So `parse()` gives us **all the important information about a path**.

path.resolve() -

It creates an absolute path. **Absolute path** means a complete path that tells you exactly where a file or folder is located, starting from the root of the file system.

For example:

```
const path = require("path");
```

```
const filePath = path.resolve("students", "student.txt");
```

```
console.log(filePath);
```

You might get:

C:\Users\Ekta\NodeProject\students\student.txt

The exact result depends on where your project is located.

Difference:

path.join("students", "student.txt")

creates:

students/student.txt

while:

path.resolve("students", "student.txt")

creates an:

absolute path

__dirname

When using **CommonJS**, Node.js provides a special variable called `__dirname`. It represents the **directory where the current JavaScript file is located**.

Suppose your project is:

```
NodeProject/  
|  
├── app.js  
└── files/  
    └── student.txt
```

Inside `app.js`:

```
console.log(__dirname);
```

You might get:

C:\Users\Ekta\NodeProject

__filename

It gives the **complete path of the current file**.

```
console.log(__filename);
```

For example:

C:\Users\Ekta\NodeProject\app.js

__dirname - Folder containing the current file

__filename - Complete path of the current file

Method	Purpose	Example
<code>path.join()</code>	Joins path pieces	<code>files/student.txt</code>
<code>path.basename()</code>	Gets filename	<code>student.txt</code>
<code>path.dirname()</code>	Gets directory	<code>/files</code>
<code>path.extname()</code>	Gets extension	<code>.txt</code>
<code>path.parse()</code>	Breaks path into pieces	<code>{root, dir, base, ext, name}</code>
<code>path.resolve()</code>	Creates absolute path	<code>C:\Project\files\student.txt</code>

fs works with the actual files and folders, while **path** helps us correctly locate and construct the paths to those files and folders.

What is a Server?

If we just look at the word itself, server is the one that serves something. The system, that is making the request for something, is called the client. Server accepts the request from the client and sends the response based on the request.

HTTP

It stands for Hyper Text Transfer Protocol . Set of Protocols or rules that are required to communicate between client and server. HTTPS:- It is exactly the same with an added layer of security, it is achieved by SSL(Secure Socket Layer).

NodeJS HTTP Module

The Node.js HTTP module is a core built-in module used to create and manage HTTP servers and handle client–server communication.

- Included using the `require()` method.
- Supports standard HTTP methods like GET, POST, PUT, and DELETE.
- Simplifies handling of requests, headers, query parameters, and responses.

```
const http = require("http")
```

Features -

Here are some features of the Node.js HTTP module:

- **Server Creation:** Enables creation of HTTP servers.
- **Request Handling:** Listens for and processes incoming HTTP requests.
- **Response Handling:** Sends responses to clients, supporting dynamic content delivery.
- **HTTP Method Support:** Supports methods such as GET, POST, PUT, DELETE, and others.

Creating Servers using HTTP -

Create a server using `http.createServer()` that listens for requests.

```
const http=require("http");

const server=http.createServer((req,res)=>{

    res.end("Hello world");

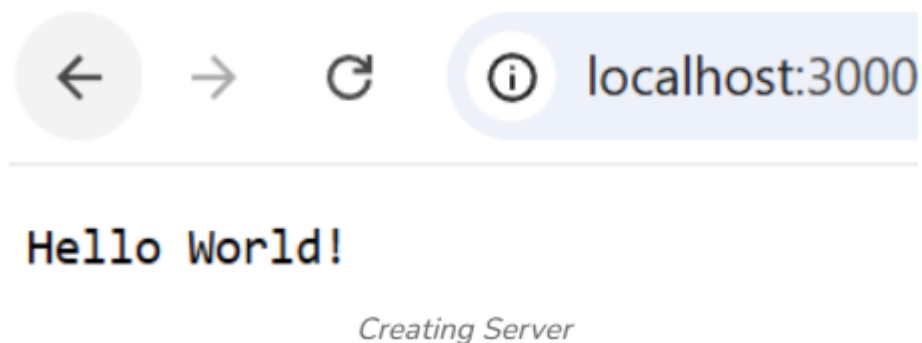
});

server.listen(8080,()=>{

console.log("Running in 8080");

});
```

Output:



Creating HTTP Server with Routing -

We can use Node.js's built-in **HTTP module** to create a server and handle different URLs.

Example -

```
const http = require("http");

const server = http.createServer((req, res) => {
  if (req.url === "/") {

    res.end("Welcome to homePage");

  } else if (req.url === "/about") {

    res.end("Welcome to About Page");

  } else if (req.url === "/cart") {

    res.end("Welcome to Cart page");

  }
});
server.listen(8080, () => {
  console.log("Running in 8080");
});
```

Managing everything with if-else becomes difficult. Express.js makes server and route handling easier.

You can find and stop the process using port 8081. In the VS Code terminal, run:

```
netstat -ano | findstr :8081
```

You may see something like:

```
TCP 0.0.0.0:8081 0.0.0.0:0 LISTENING 12345
```

The last number is 12345. It is the **PID (Process ID)**.

Then run:

```
taskkill /PID 12345 /F
```

Replace 12345 with the PID you actually get. Then start your server again:

```
node server.js
```

Creating a Server using Express.js -

- **Express.js** is a popular framework for building **web applications and APIs** using Node.js.
- It makes it easier to create and manage a server compared to using Node.js's built-in **http** module.
- When we create an Express.js application, we set up the **foundation for handling incoming requests** and defining how our application **responds to those requests**.

Analogy -

Assume a **server as a waiter in a restaurant**:

- **Client** → Customer
- **Request** → Customer places an order
- **Server** → Waiter
- **Response** → Waiter brings the requested food

In Node.js, we create a server so that it can receive requests and send responses.

What is **localhost**?

- **localhost** refers to your own computer.
- When we run a server on our computer, we can access it through **localhost**.
- For example:

<http://localhost:3000>

Here:

- **localhost** → Your own computer
- **3000** → Port on which the server is running

Example: If your Express server is running on port **3000**, opening <http://localhost:3000> in the browser sends a request to the server running on your computer.

```
const express=require('express')  
  
const app=express()
```

```
app.get("/", (req, res) => {  
    res.send("Welcome to home page")  
})  
  
app.listen(8080, () => console.log("Server started in  
http://localhost:8080"))
```

- Express is imported using **require('express')**, and an app instance is created with **express()**.
- A route is defined using the **app.get()** method, which responds with a message when the root URL (/) is accessed.
- The **app.listen()** method starts the server and listens on port 8080 for incoming requests.

CRUD Application Using Express.js -

1. Creating a CRUD Application -

Till now, we have learned how to create an Express server, create routes, handle requests and responses, and use middleware.

Now, we will combine these concepts and create a simple **CRUD Application using Express.js**.

CRUD stands for:

- **C** → **Create** → Add new data
- **R** → **Read** → Get existing data
- **U** → **Update** → Modify existing data
- **D** → **Delete** → Remove data

For our application, we will manage **student data**.

For example:

Student

id

name
age
course

We will gradually create APIs to:

Create → Add a student
Read → Get students
Update → Update a student
Delete → Delete a student

We will first create a **basic Express application**, then create the **GET request**, and finally create the **POST request**.

Part 1: Create a Basic Express Application

Before creating CRUD APIs, let's first create a basic Express application.

Step 1: Create a Project -

Create a new folder:

```
express-crud
```

Open the folder in VS Code.

Initialize a Node.js project:

```
npm init -y
```

Install Express:

```
npm install express
```

Step 2: Create **server.js** -

Create a file: **server.js**

```
const express = require("express"); //Import Express framework  
  
const app = express(); //Create Express application
```

```
app.listen(8080, () => {  
  console.log("Server is running on  
http://localhost:8080/");  
});
```

Explanation -

1. Import Express -

```
const express = require("express");
```

express is a Node.js framework used to create web servers and APIs easily.

Here, **require("express")** loads the Express package.

We store it inside the **express** variable.

So we can use Express in our application.

2. Create Express Application -

```
const app = express();
```

express() creates an Express application. We store that application inside the **app** variable. We will use an app to create routes and configure our server.

For example:

```
app.get(...)  
app.post(...)  
app.use(...)  
app.listen(...)
```

3. Start the Server -

```
app.listen(8080, () => {  
  console.log("Server is running on http://localhost:8080/");  
});
```

`app.listen()` starts our Express server. Here, 8080 is the port number. So our application will run at: <http://localhost:8080/>

The function:

```
() => {  
  console.log("Server is running...");  
}
```

is a **callback function**. It runs when the server successfully starts.

Part 2: Create Our First GET Request -

Now that our basic Express server is ready, let's create a simple GET request.

Add the following code:

```
const express = require("express");  
  
const app = express();  
  
// GET request  
  
app.get("/", (req, res) => {  
  
  res.send("Welcome to Express Application");  
  
});  
  
app.listen(8080, () => {  
  
  console.log("Server is running on  
http://localhost:8080/");  
  
});
```

Explanation of GET Request -

```
app.get("/", (req, res) => {  
  
  res.send("Welcome to Express Application");  
  
});
```

app.get() is used to create a **GET route**. The GET request is generally used when the client wants to **read or retrieve data** from the server. The **"/"** represents the home route. So when the client sends:

GET http://localhost:8080/

this route will execute.

In (req, res), **req** stands for **request**. It contains information coming from the client.

For example:

- URL
- parameters
- query data
- request body
- headers

res stands for **response**. It is used by the server to send a response back to the client. For example:

```
res.send("Hello");
```

or:

```
res.json(data);
```

```
res.send("Welcome to Express Application");
```

res.send() sends a response to the client. In this example, the browser will display:

Welcome to Express Application

Part 3: Create a Student Data File -

Now let's start building our CRUD application. Create a file: students.json

Add some student data:

```
[  
  
  {
```

```
    "id": 1,

    "name": "Aman"

  },

  {

    "id": 2,

    "name": "Yogesh"

  },

  {

    "id": 3,

    "name": "Rahul"

  }

]
```

Our project structure will now look like:

```
express-crud
|
├─ node_modules
├─ students.json
├─ package.json
├─ package-lock.json
└─ server.js
```

Part 4: Import the File System Module -

We need to read and write data inside **students.json**. For this, Node.js provides the **File System (fs)** module.

Add:

```
const fs=require("fs")
```

So our code becomes:

```
const express = require("express");

const fs = require("fs");

const app = express();
```

Why do we need **fs**?

The **fs** module allows Node.js to work with files.

We can use it to:

- Read files
- Create files
- Write files
- Update files
- Delete files

In our CRUD application, we will use it to store student data in: students.json

Part 5: Enable JSON Data from Client -

Before creating our POST API, we need to understand one important middleware.

Add:

```
app.use(express.json());
```

So:

```
const express = require("express");
```

```
const fs = require("fs");
```

```
const app = express();
```

```
app.use(express.json());
```

What does **express.json()** do?

When a client sends JSON data in the request body, Express needs to understand that JSON data. For example, the client may send:

```
{
```

```
"name": "Aman",  
"age": 20  
}
```

The middleware: **app.use(express.json());**

parses the JSON request body and makes it available through: **req.body**

For example:

```
console.log(req.body);
```

can give:

```
{  
  name: "Aman",  
  age: 20  
}
```

Without `express.json()`, Express will not automatically parse incoming JSON request bodies.

Part 6: Create GET API to Read Students -

Now let's create our first actual CRUD API. We want:

GET /students

to return all students.

Add:

```
app.get("/students", (req, res) => {  
  
  const data = fs.readFileSync("students.json", "utf-8");  
  
  const students = JSON.parse(data);
```

```
res.send(students);  
});
```

Explanation of GET API -

Step 1: Create the route

```
app.get("/students", (req, res) => {
```

This creates a GET API:

GET /students

When the client requests:

http://localhost:8080/students

this function will execute.

Step 2: Read the JSON file -

```
const data = fs.readFileSync("students.json", "utf-8");
```

fs.readFileSync() reads the contents of the file. We are reading: students.json. The "utf-8" tells Node.js to convert the file data into readable text. The result is stored in data variable. For example: data may contain:

```
[  
  {  
    "id": 1,  
    "name": "Aman"  
  },  
  {  
    "id": 2,  
    "name": "Yogesh"  
  }  
]
```

```
]
```

At this point, `data` is a **string**.

Step 3: Convert JSON string into JavaScript data -

```
const students = JSON.parse(data);
```

JSON.parse() converts JSON-formatted text into a JavaScript value. So:

JSON string

↓

JSON.parse()

↓

JavaScript array/object

Now `students` is a JavaScript array.

Step 4: Send the students -

```
res.send(students);
```

This sends the student data back to the client. The response will look like:

```
[  
  {  
    "id": 1,  
    "name": "Aman"  
  },  
  {  
    "id": 2,  
    "name": "Yogesh"  
  },  
  {
```

```
"id": 3,  
  "name": "Rahul"  
}  
]
```

Part 7: Create POST API to Add a Student -

We have completed the **Read** operation. Now let's implement the **Create** operation. We want the client to send a new student:

```
{  
  "name": "Vikas"  
}
```

and add it to `students.json`.

Create the following POST API:

```
app.post("/students", (req, res) => {  
  
  const data = fs.readFileSync("students.json",  
    "utf-8");  
  
  const students = JSON.parse(data);  
  
  console.log(students);  
  
  const newStudent = {  
    "id": students.length + 1,  
    ...req.body
```

```
};

students.push(newStudent);

fs.writeFileSync(

  "students.json",

  JSON.stringify(students)

);

res.send("Student Submitted Successfully");

});
```

Explanation of POST API -

Step 1: Create POST Route -

```
app.post("/students", (req, res) => {
```

This creates a POST API:

POST /students

POST is generally used when we want to **create/add new data**.

Step 2: Read Existing Students -

```
const data = fs.readFileSync("students.json", "utf-8");
```

We first read the existing student data.

Why?

Because we don't want to overwrite the existing students. We want to add a new student to them.

Step 3: Convert JSON String -

```
const students = JSON.parse(data);
```

The file data is initially a string. `JSON.parse()` converts it into a JavaScript array.
For example:

```
[  
  { id: 1, name: "Aman" },  
  { id: 2, name: "Yogesh" }  
]
```

Step 4: Get Data from Client -

`req.body`

contains the JSON data sent by the client.

For example, the client sends:

```
{  
  "name": "Vikas"  
}
```

Then:

`req.body`

will contain:

```
{  
  name: "Vikas"  
}
```

This works because we already added: `app.use(express.json());`

Step 5: Create New Student -

```
const newStudent = {
```

```
"id": students.length + 1,  
...req.body  
};
```

Suppose we already have 3 students. Then:

students.length is:

3

Therefore:

students.length + 1 is:

4

So the new student becomes:

```
{  
  "id": 4,  
  "name": "Vikas"  
}
```

What does `...req.body` mean?

The `...` is the **spread operator**. Suppose req.body is:

```
{  
  "name": "Vikas"  
}
```

Then:

```
{  
  "id": 4,  
  ...req.body  
}
```

becomes:

```
{  
  "id": 4,  
  "name": "Vikas"  
}
```

So we are combining: id + client data = new student

Step 6: Add Student to Array -

```
students.push(newStudent);
```

`push()` adds the new student to the existing array. Before:

```
[  
  { id: 1, name: "Aman" },  
  { id: 2, name: "Yogesh" },  
  { id: 3, name: "Rahul" }  
]
```

After:

```
[  
  { id: 1, name: "Aman" },  
  { id: 2, name: "Yogesh" },  
  { id: 3, name: "Rahul" },  
  { id: 4, name: "Vikas" }  
]
```

Step 7: Save Updated Data -

```
fs.writeFileSync(  
  "students.json",
```

```
JSON.stringify(students)

);
```

After adding the new student to the array, we need to save the updated array back into the file.

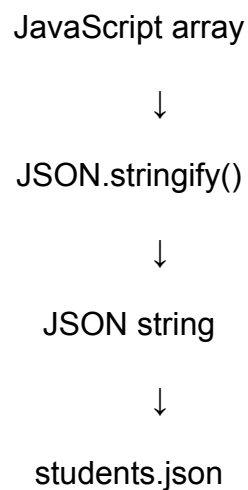
`fs.writeFileSync()` writes data into a file.

Why did we use `JSON.stringify()`?

Our `students` variable is a JavaScript array:

```
[
  { id: 1, name: "Aman" },
  { id: 2, name: "Yogesh" }
]
```

But the file stores JSON text. Therefore, we convert the JavaScript array back into JSON text:



Step 8: Send Response -

```
res.send("Student Submitted Successfully");
```

After successfully adding the student, we send a response to the client. The client receives:

Student Submitted Successfully

Complete Application So Far -

At this point, our application supports:

CREATE → POST /students

READ → GET /students

Complete code:

```
const express = require("express");    // Import Express
framework

const fs = require("fs");                // Import File
System module

const app = express();                  // Create Express
application

// Middleware

// Used to read JSON data sent by the client

app.use(express.json());

// GET /

// Basic route

app.get("/", (req, res) => {

    res.send("Welcome to Express Application");

});
```

```
// GET /students

// Read all students

app.get("/students", (req, res) => {

    // Read students.json file

    const data = fs.readFileSync("students.json",
"utf-8");

    // Convert JSON string into JavaScript array

    const students = JSON.parse(data);

    // Send students to client

    res.send(students);

});

// POST /students

// Add a new student

app.post("/students", (req, res) => {

    // Read existing students

    const data = fs.readFileSync("students.json",
"utf-8");
```

```
// Convert JSON string into JavaScript array

const students = JSON.parse(data);

// Create new student

const newStudent = {

  id: students.length + 1,

  ...req.body

};

// Add new student to array

students.push(newStudent);

// Convert array to JSON and save in file

fs.writeFileSync(

  "students.json",

  JSON.stringify(students)

);

// Send response

res.send("Student Submitted Successfully");

});
```

```
// Start server

app.listen(8080, () => {

    console.log("Server is running on
http://localhost:8080/");

});
```